



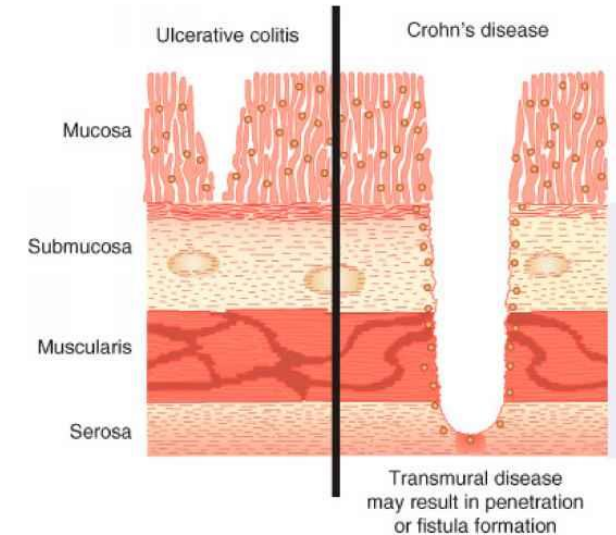
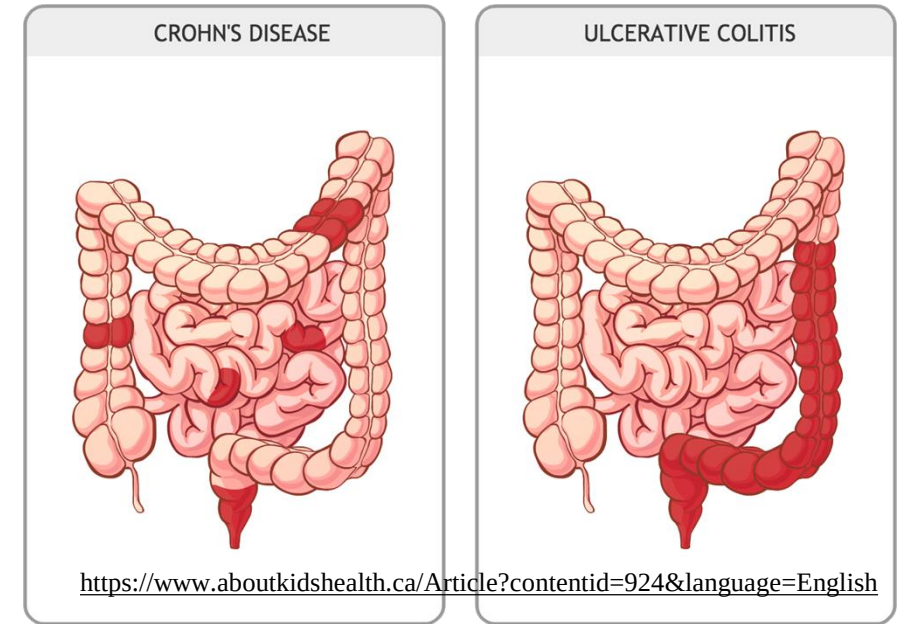
Inflammatory bowel disease (IBD)

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ภาควิชาเภสัชวิทยา คณะเภสัชศาสตร์ มหาวิทยาลัยมหิดล

Inflammatory Bowel Disease (IBD)

- **Definition:**
 - Chronic condition resulting from inappropriate mucosal immune activation.
- **Subtype**
 - **Crohn's disease (CD)**
 - Sex: more common in female
 - Variable age of onset
 - Symptoms: depend on location of disease
 - Abdominal pain, diarrhea, weight loss, fatigue
 - Bloody stool: variable
 - Malnutrition: common
 - **Ulcerative colitis (UC);** mucosa inflammation
 - 90% relapse in first 2 years
 - Sex: more common in female
 - Age of onset: 15-35 yrs. and 55-70 yrs.
 - Symptoms:
 - Stool urgency, fatigue, increase bowel movements, mucous in stool, nocturnal bowel movements and abdominal pain
 - Bloody stool: common
 - Malnutrition: less common
 - 90% relapse in first 2 years

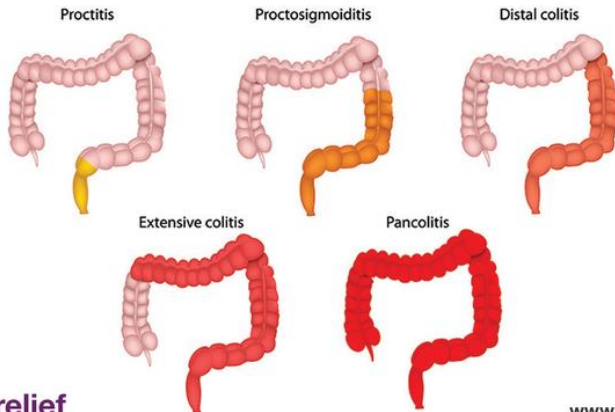


IBD: Subtypes

Ulcerative colitis

- Proctitis (rectum)
- Proctosigmoiditis (rectum + sigmoid)
- Distal colitis (left sided)
- Extensive colitis
- Pancolitis (entry part of colon)

TYPES OF ULCERATIVE COLITIS



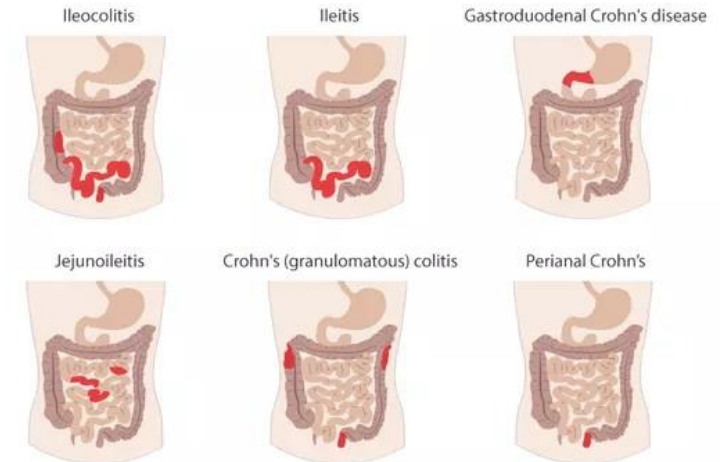
IBDrelief

www.ibdrelief.com

Crohn's disease

- Ileocolitis
- Ileitis
- Gastroduodenal Crohn's disease
- Jejunoileitis
- Crohn's (granulomatous) colitis
- Perianal Crohn's

TYPES OF CROHN'S DISEASE



IBDrelief

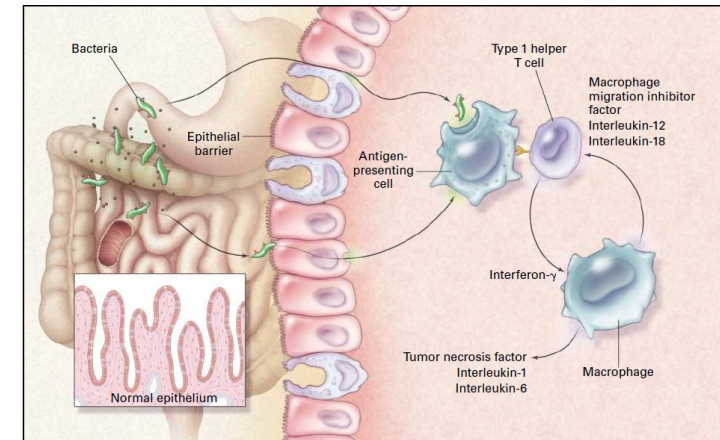
www.ibdrelief.com

IBD: Clinical features

Clinical presentation	Ulcerative colitis	Crohn's disease
General	- May present with similar symptoms - Onset: insidious or subacute - May occur extraintestinal manifestation - Indeterminate colitis (can not distinguish between UC and CD); approx. 10%	
Signs	fever, tachycardia (severe), dehydration, arthritis, hemorrhoids, anal fissures, perirectal abscesses	fever, tachycardia (severe), dehydration, arthritis, abdominal mass and tenderness, perianal fissure or fistula
Symptoms	diarrhea(bloody, watery, or mucopurulent), rectal bleeding, abdominal pain/cramping, weight loss and malnutrition, tenesmus , constipation(with proctitis)	diarrhea(less bloody than UC), rectal bleeding(less than UC), abdominal pain/cramping, weight loss and malnutrition (more common than UC), fatigue/malaise
Laboratory tests	leukocytosis, decreased hematocrit/hemoglobin, elevated erythrocyte sedimentation rate(ESR) or C-reactive protein (CRP), positive test for fecal occult blood, (+) perinuclear antineutrophil cytoplasmic antibodies (pANCA; 70% of cases), Clostridium difficile toxin assay	leukocytosis, decreased hematocrit/hemoglobin, elevated erythrocyte sedimentation rate(ESR) or C-reactive protein (CRP), positive test for fecal occult blood, (+) anti-Saccharomyces cerevisiae antibodies (50% of cases), hypoalbuminemia (severe)

IBD: Pathogenesis

- **Pathogenesis:**
 - Idiopathic disorder
 - Not autoimmune disease
- **Etiology**
 - Currently considered as an inappropriate immune response to the endogenous microbiota within the intestines
- **Factors**
 - Defective immune regulation
 - Genetic mutation
 - Exogenous factors



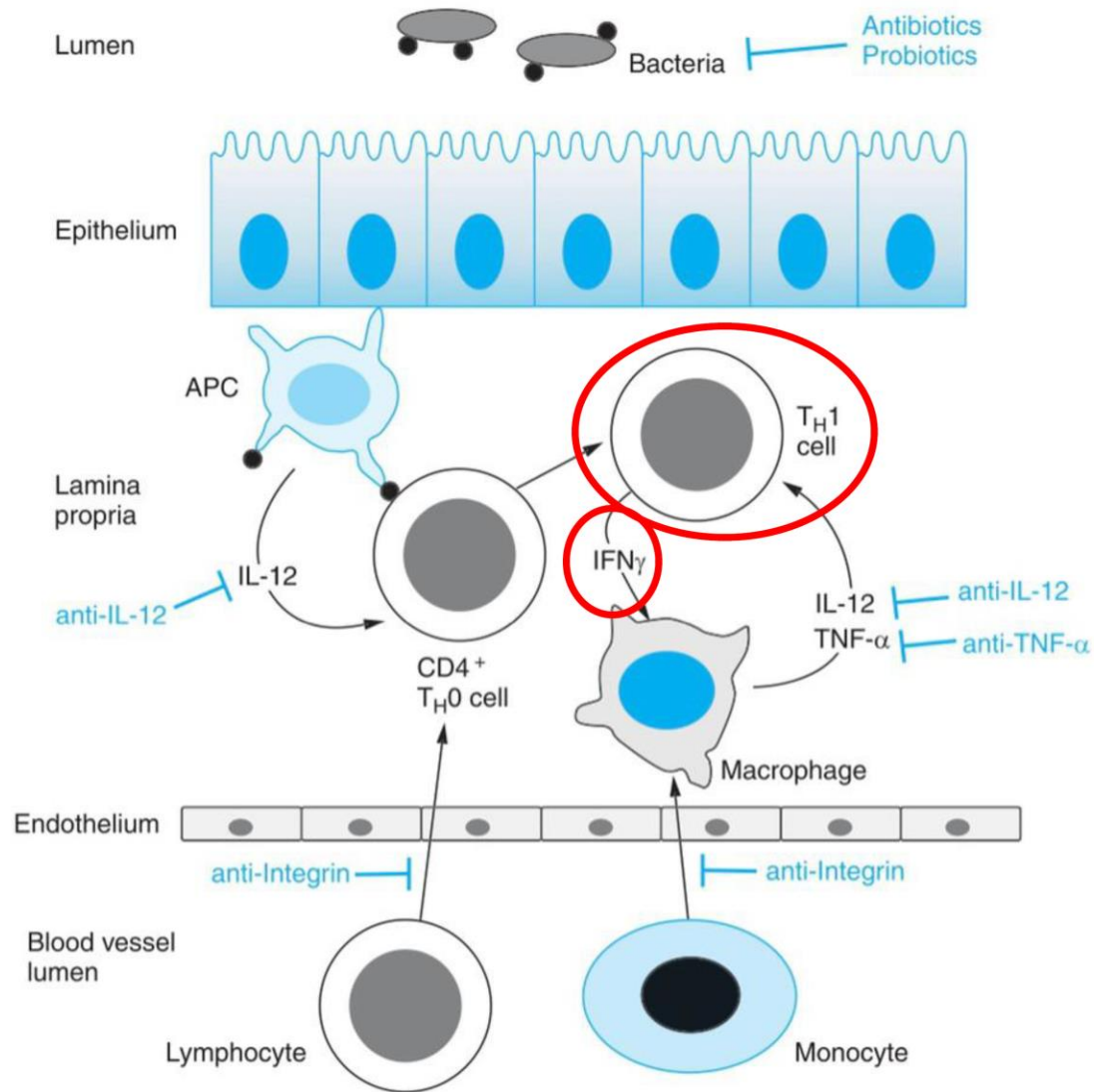
N Engl J Med, august 8, 2002: 347(6)

Pathogenesis of Inflammatory Bowel Disease.

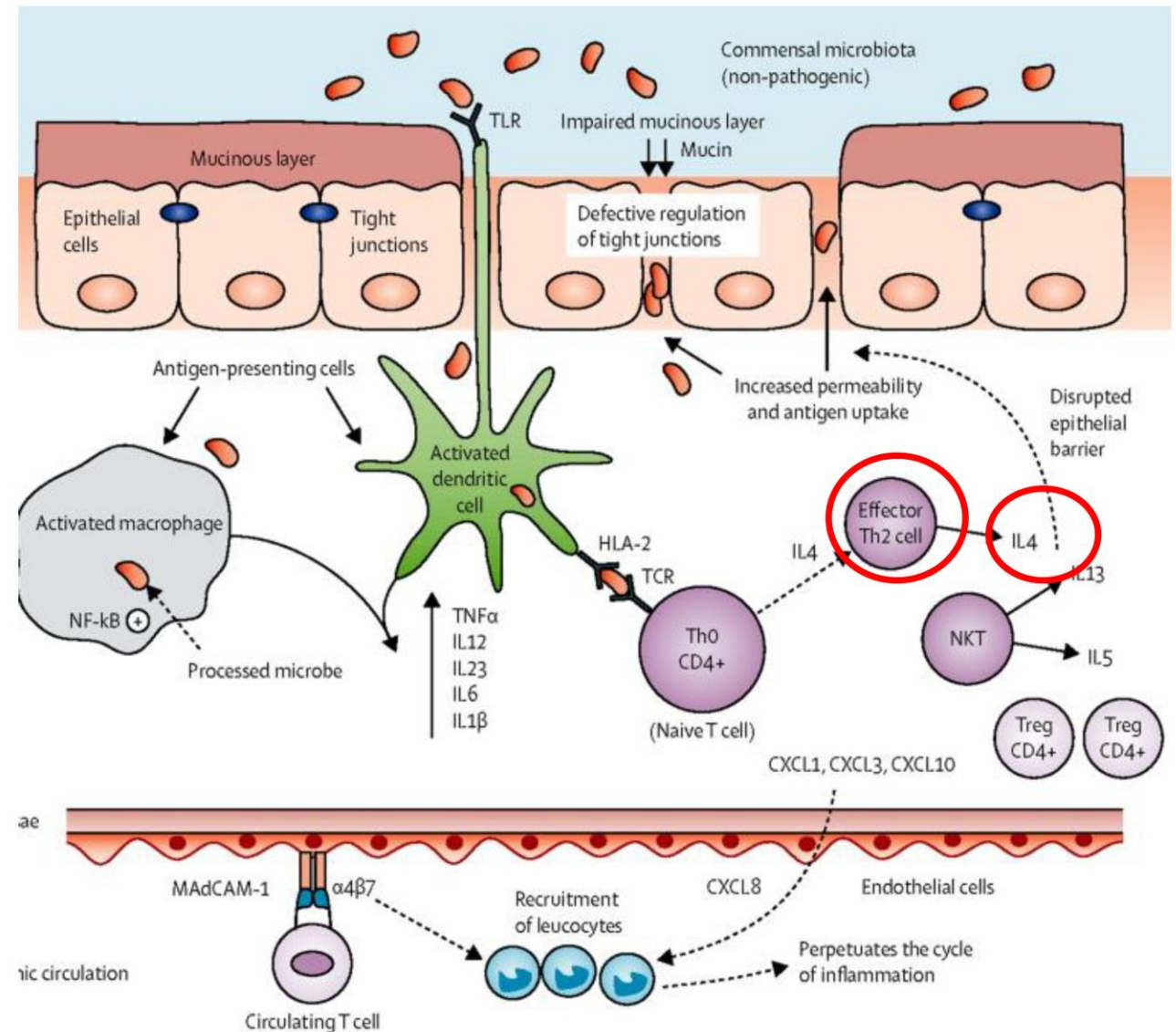
- Normal epithelium have tight junctions and goblet-cell populations that produce peptides and mucin glycoproteins to provide a barrier against luminal agents.
- Intestinal bacterial or antigen from dietary may stimulate mucosal immune system result in chronic intestinal inflammation.
 - Bacterial products penetrate through the mucosal barrier, then activate adaptive immune response.
 - Bacterial products stimulate the surface epithelium (innate immune receptor), then promote the production of cytokines and chemokines that recruit and activate mucosal immune cells.
- Promote differentiation of immune cells.
 - Crohn's disease: type 1 helper T cells (Th1), and Th17
 - Interferon- γ , IL-1, IL-6, IL-8, IL-12, TNF- α
 - Ulcerative colitis: atypical type 2 helper T cells, and Th17
 - IL-1, IL-6, IL-13, TNF- α
 - TNF- α is major contribution to inflammatory process
 - Promote other cell type
 - Macrophage
 - Vascular endothelium: procoagulant effects
 - Mucosal cells: production of matrix metalloproteinase
 - Cellular stimulation
 - Production of NF- κ B which stimulates the production of proinflammatory cytokines

IBD: Pathogenesis

Crohn's disease

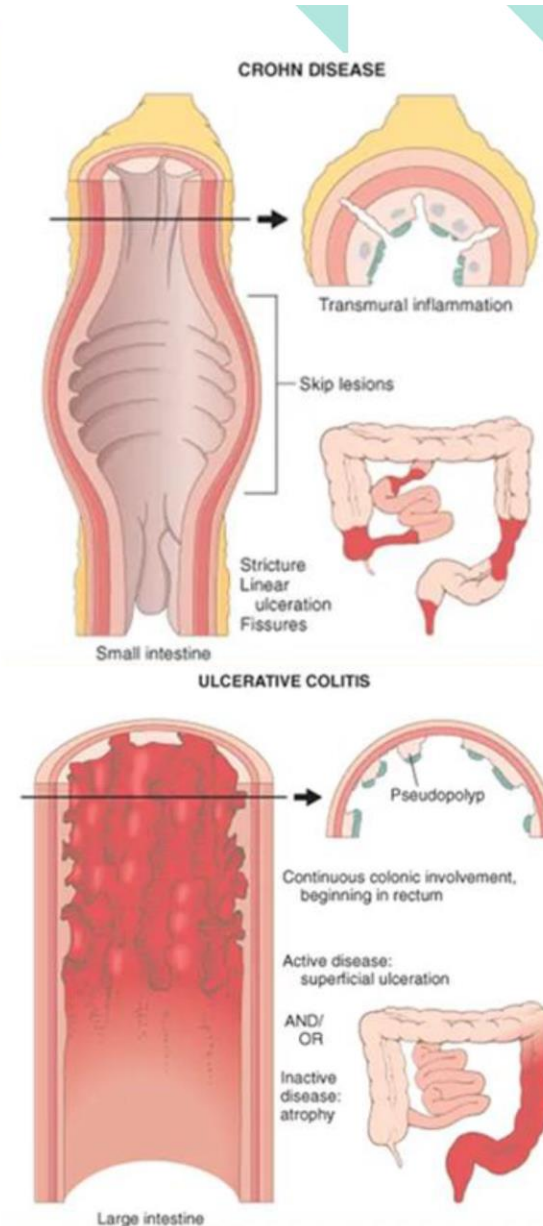


Ulcerative colitis



IBD: Pathogenesis

	Crohn's disease	Ulcerative colitis
Bowel region	GI tract most commonly the area adjacent to ileocecal valve	Large Intestine only starting at the anal verge and extending proximally
Distribution	Segmental with skip areas	Continuous without skip areas
Inflammation	Transmural	Limited to mucosal
Helper T cell	helper T cell-1	helper T cell-2
Important cytokines	TNF-α IFN-γ	IL-4 IL-13 TNF-α
Additional symptom	Anal tears, Ulcers, Infections, fistulas and intestinal obstruction	Rectal bleeding or blood in stools more often than CD



IBD: Etiology

- **Defective immune regulation**
- **Genetic factors: family history is a strong predictor of IBD**; more correlated in CD.
 - **Mutation/polymorphism**
 - NOD2 (nucleotide oligomerization binding domain2)
 - CD; Caucasian (<10% of cases)
 - Bind to intracellular bacterial peptidoglycan, and activates NF- κ B
 - **ATG16L1 (autophagy-related 16-like)**
 - **IRGM (immunity-related GTPase M)**
 - Mutation of genes that shared with UC and CD
 - JAK2
 - IL-23R (protective)-involve in Th17 maturation
 - Some genetic loci are associated with increasing risk of other diseases (rheumatoid arthritis, psoriasis)

- **Exogenous factors**
 - IBD may be triggered by microbe
 - Diagnosis by the presence of antibodies against them (E.coli OMP2 antibody, ASCA)
 - Body may react against normal flora
 - Antibiotics may improve the symptoms
 - Psychosocial factors (major lifestyle, daily stress)
 - Cause worsening of symptoms
 - Drugs
 - NSAIDs: disrupt the integrity of GI mucosa may promote IBD
 - Contraceptive: some cohort studies
 - Smoking
 - Protective effect in UC
 - Worsening in CD

IBD: treatment

Aims of treatment

- Acute episode
 - alleviation of signs and symptoms both clinically and endoscopically
 - Suppression of inflammatory
- Remission
 - Maintenance remission
 - Prevent disease relapse
 - Prevent the need for surgical intervention or hospitalization
 - Manage extraintestinal manifestation
 - Prevent malnutrition
 - Prevent treatment-associated ADR
- Improve quality of life

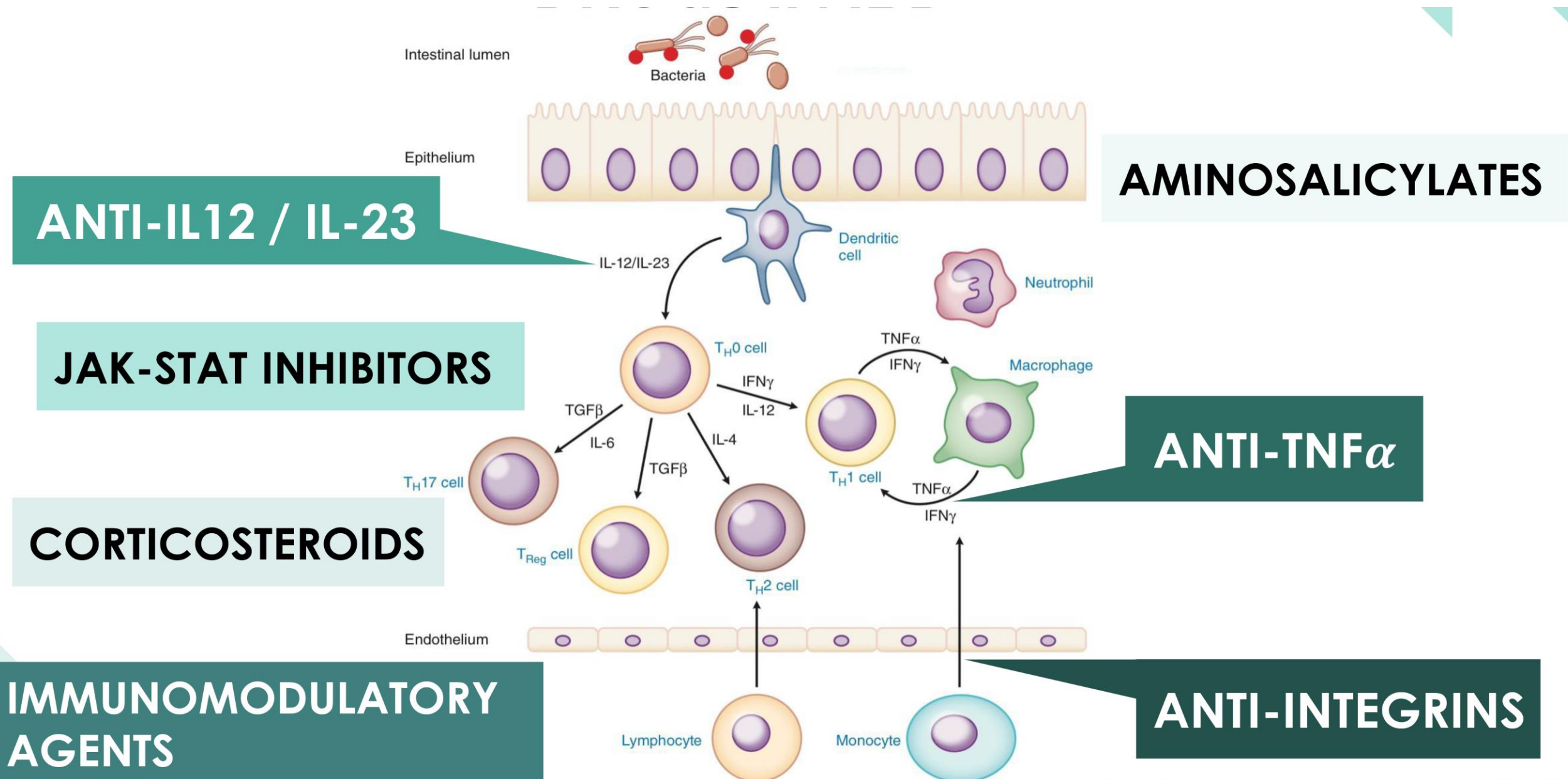
Pharmacological treatment

- 5-ASA based therapy
- Glucocorticoids
- Immunosuppressants
- Immunomodulators

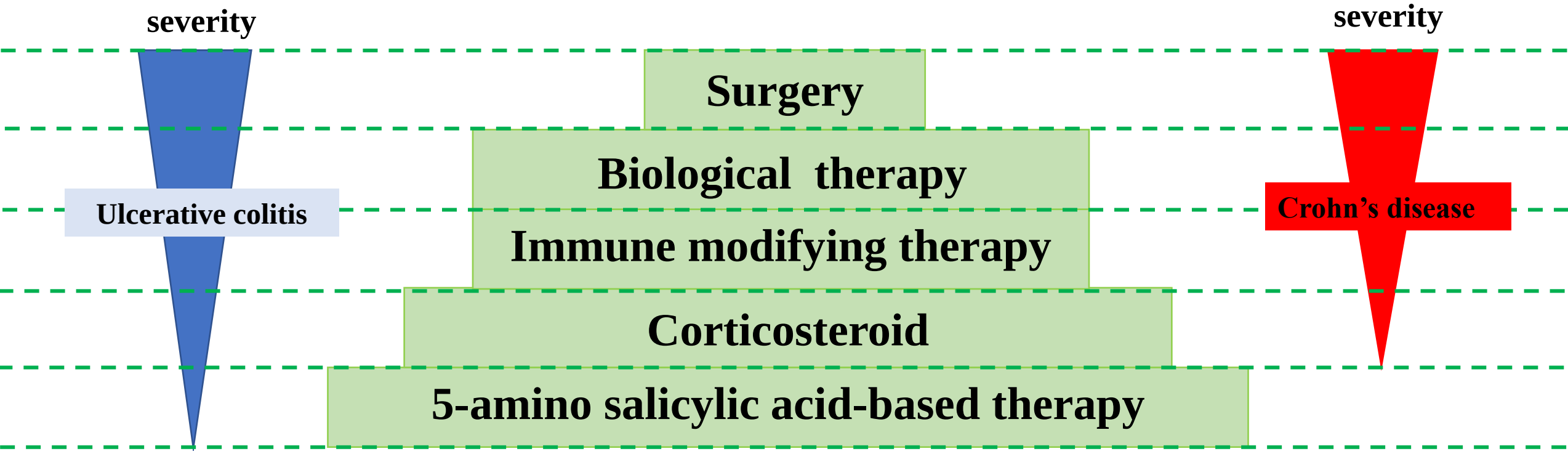
Non-Pharmacological treatment

- Education
- Nutrition

IBD: treatment



IBD: treatment pyramid



IBD: clinical severity

Severity of IBD	Mild	Moderate	severe
Number of bloody stools/day	<4	4-6	>6
Temperature (°C)	Afebrile	Intermediate	>37.8
Pulse (beats per min)	Normal	Intermediate	>90
Hemoglobin (g/L)	>110	105-110	<105
Erythrocyte sedimentation rate (mm/h)	<20	20-30	>30

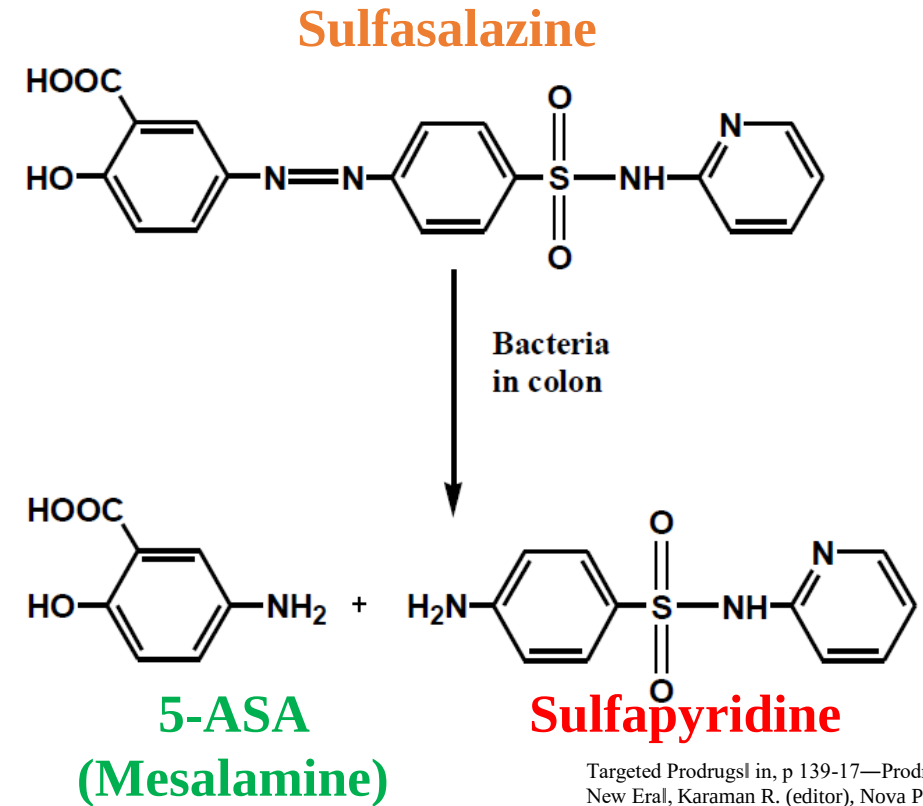
IBD: 5-ASA

5-Aminosalicylate (mesalamine) based therapy

- First-line therapy for mild to moderate UC (*1), **not in CD(*2)**
- Effective in induction remission of UC, **not in CD**
- Effective in maintenance of remission in UC

Mechanisms of action

- No related to COX inhibition
- Various mechanisms
 - Inhibiting IL-1 and TNF-alpha production
 - Inhibiting Lipo-oxygenase pathway
 - Scavenging of free radicals
 - Inhibiting of NF- κ b via PPAR-gamma



Targeted Prodrugs in, p 139-17—Prodrugs Design – A
New Eral, Karaman R. (editor), Nova Publisher, USA,

- Azo bond prevents absorption in stomach and small bowel
- Broken down in colon by bacterial azo reductase

Sulfasalazine: Pharmacokinetics/ADR

- 20-30% absorbed in small intestine, and is excreted unchanged
- 70% reaches the colon
- Release to give
 - 5-ASA
 - 400 mg from 1 g of parent compound
 - Sulfapyridine
 - Highly lipid soluble
 - Completely absorbed
 - Undergoes acetylation and glutathione conjugation
 - Renally excretion

- ADR primarily due to sulfapyridine
- Dose related
 - headache, fatigue, nausea
- Allergic reaction
 - Rash, Steven-Johnson syndrome, Hepatitis, Bone marrow suppression
- Decrease in sperm number and motility
 - reversible
- Nephrotoxicity (1:4000 patient years). (1,2)
 - Monitor renal base-line (R/O IBD-associated nephrotoxicity).
 - Repeat after 2-3 months
 - Median time before renal impairment; 3 years (95% CI 2.3-3.7). (3)
- Treatment
 - Drug withdraw
 - High-dose oral corticosteroid

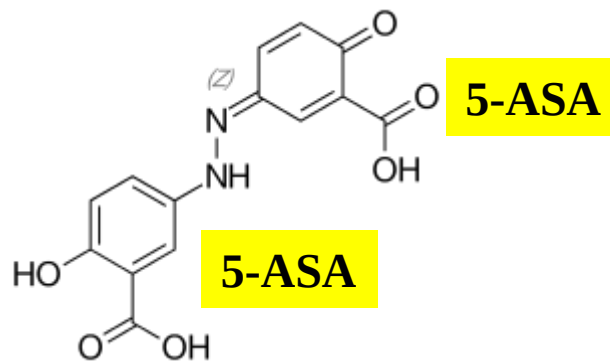
1. Muller AF, Stevens PE, McIntyre AS, et al. Experience of 5-aminosalicylate nephrotoxicity in the United Kingdom. *Aliment Pharmacol Ther* 2005;21:1217-24.

2. Gisbert JP, Luna M, Gonzalez-Lama Y, et al. Effect of 5-aminosalicylates on renal function in patients with inflammatory bowel disease: 4-year follow-up study. *Gastroenterol Hepatol* 2008;31:477-84.

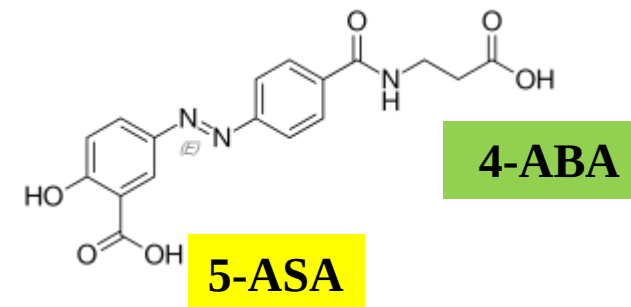
3. Heap GA, So K, Weedon M, et al. Clinical features and HLA association of 5-aminosalicylate (5-ASA)-induced nephrotoxicity in inflammatory bowel disease. *J Crohns Colitis* 2016;10:149-58.

Newer Generation of 5-ASA

- Contain the same Azo bound, replaced sulfapyridine by other compound
- Can be used safely in patients with a reported sulfonamide allergy
 - **Olsalazine**
 - 2 mesalamine molecules
 - ADR: high incidence of secretory diarrhea
 - **Balsalazide**
 - Mesalamine
 - 4-aminobenzoyl- β -alanine (4-ABA); inert carrier
- Higher doses can be used with less side effects



Olsalazine

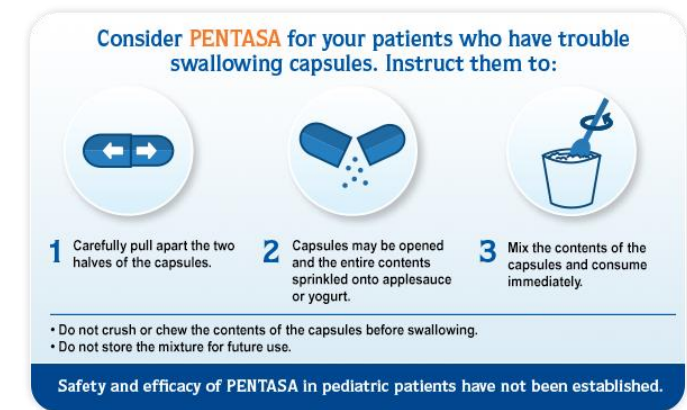


Balsalazide

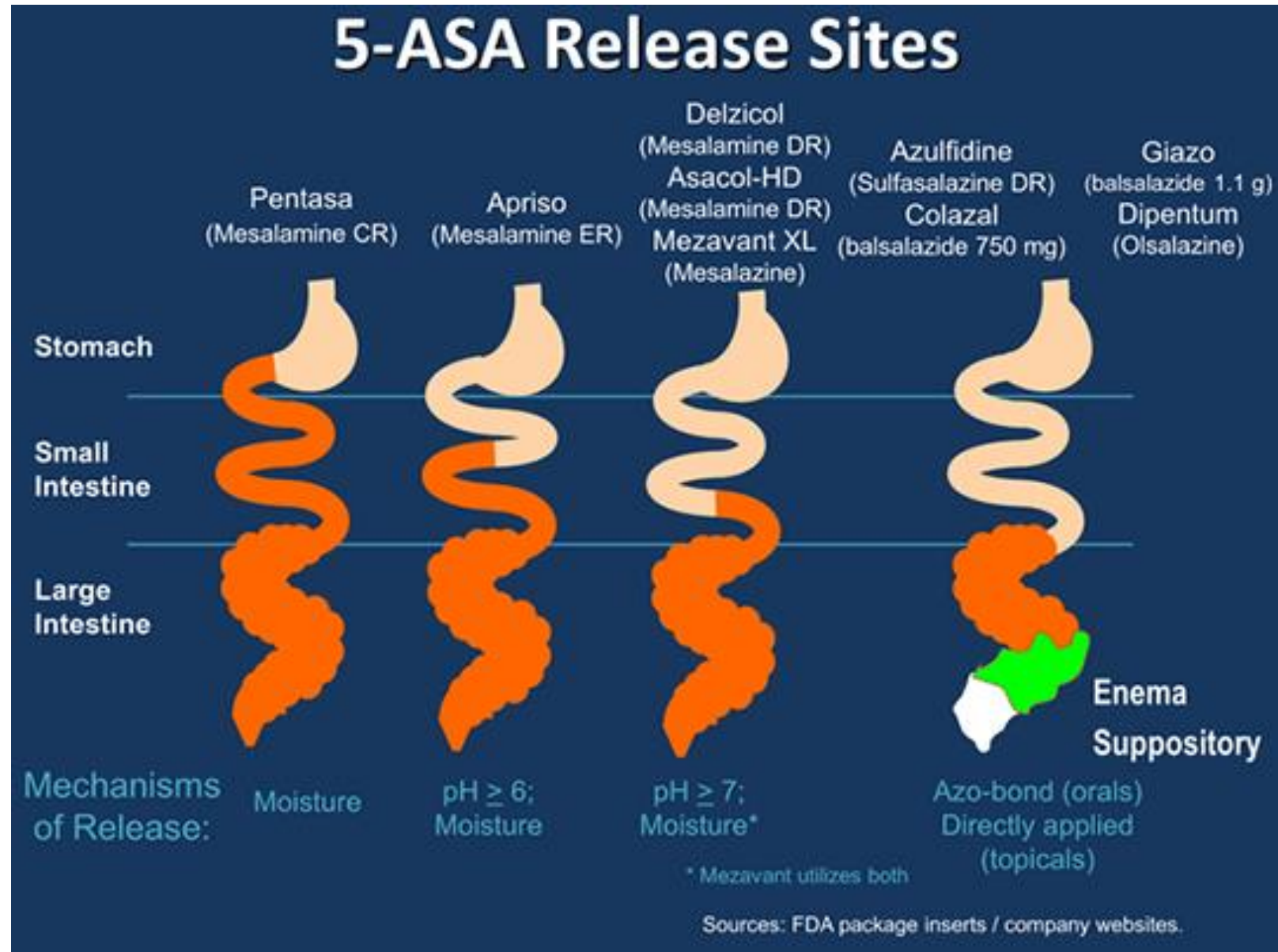
5-ASA: Coated drugs

- **pH sensitive release tablet (Asacol®, Apriso®)**
 - Enteric-coated form of **mesalamine**
 - Do not crush, chew, or break
 - Liberated at pH>7
 - Site of release: **terminal ileum and colon**
 - Dose:
 - Once-daily
 - 2.4-4.8g/day (acute)
 - 1.6-4.8 g/day (maintenance)
- **pH sensitive release tablet (Lialda®)**
 - MMX: multi-matrix technology of **mesalamine**
 - Do not crush, chew, or break
 - Encloses 5-ASA inside a lipophilic matrix within a hydrophilic matrix
 - Further encapsulated by a pH-sensitive polymer which release drug at PH>7
 - Dose:
 - Once-daily
 - 2.4-4.8 g/day

- **Pentasa®**
 - Sustained release formulation
 - Do not crush, chew, or break
 - Disintegration of capsule occurs in the stomach
 - Ethylcellulose coating to allow water absorption into the small beads containing the **mesalamine**
 - Water dissolves the 5-ASA , diffused out of the bead into the lumen



Site of action of 5-ASA formulations



5-ASA:UC-active

- Standard therapy for **mild to moderate** active UC
- Dose:
 - 2-3 g/day
 - Once daily is as effective as divided dose
 - Higher dose can be used in more severe
- ADR:
 - Nephrotoxicity (sulfapyridine)
 - Monitor renal function
 - Baseline and during treatment (q 2-3 months)

- ASCEND I trials (301 patient) (1)
 - randomized 2.4 g or 4.8 g of 5-ASA (Asacol) to mild-moderate UC
 - At week 6, a similar proportion of patient's improvement (51% vs 56%, p=NS).
 - patients with moderate disease had better response to 4.8 g/day,
 - Patients with mild disease was no difference in dose-response
- ASCEND II (2)
 - Confirm with moderately UC had a better response to 4.8 g/day than 2.4 g/day (72% vs 59%, p=0.036).
- ASCEND III (772 patient with moderate UC) (3)
 - significant difference in remission, with 43% of patients on 4.8 g/day versus 35% of those on 2.4 g/day achieving remission at 6 weeks.
 - no difference in the primary end-point of treatment success (complete clinical remission or partial response)
- 5-ASA MMX trials
 - 4.8 g/day was more effective than 2.4 g/day in the subpopulation with prior exposure to 5-ASA,
 - who had an incomplete response to 8 weeks of 5-ASA MMX at 2.4 g/day
- Remission rate
 - Week 2: 10-30% of patients
 - Week 4: 30-45% of patients
 - Week 8: 35-50% of patients

5-ASA:UC-active:oral and enema

- Recommended
 - All UC patient (all severity) should be offered a combination of oral and enema 5-ASA
 - Even in pancolitis, combination is superior than monotherapy
 - incomplete response to oral 5-ASA should have topical therapy
 - A meta-analysis of four studies in active UC
 - Show a relative risk of no remission of 0.65 (95% CI 0.47 to 0.91) for combined oral and topical therapy over oral 5-ASA alone. (1)
 - A small study; 18 UC patients with frequent relapses on standard dose oral 5-ASA showed the benefits of high-dose 5-ASA combined with 5-ASA enemas. (2)
- Poor compliance; support and education are needed

5-ASA:UC-maintenance

- **5-ASA maintenance therapy decrease the risk and frequency of flares**
- **High dose 5-ASA**
 - **For who required 2 or more corticosteroid in the past year**
 - **For who become corticosteroid-dependent or refractory**
 - **For who require escalation with thiopurine**
- **Dose:**
 - Orally: 2g or more/day (1)
 - once daily dosing is as effective as divided dosing regimens (2-5)
 - 5-ASA adherence is a major problem
 - Rectal 5-ASA: distal UC

1. Wang Y, Parker CE, Feagan BG, et al. Oral 5-aminosalicylic acid for maintenance of remission in ulcerative colitis. Cochrane Database Syst Rev 2016;CD000544.
2. Li W, Zhang Z-M, Jiang X-L. Once daily vs multiple daily mesalamine therapy for mild to moderate ulcerative colitis: a meta-analysis. Colorectal Dis 2016;18:O214–23.
3. Feagan BG, MacDonald JK. Once daily oral mesalamine compared to conventional dosing for induction and maintenance of remission in ulcerative colitis: a systematic review and meta-analysis. Inflamm Bowel Dis 2012;18:1785–94.
4. Tong JL, Huang ML, Xu XT, et al. Once-daily versus multiple-daily mesalamine for patients with ulcerative colitis: a meta-analysis. J Dig Dis 2012;13:200–7.
5. Ford AC, Khan KJ, Sandborn WJ, et al. Once-daily dosing vs. conventional dosing schedule of mesalamine and relapse of quiescent ulcerative colitis: systematic review and meta-analysis. Am J Gastroenterol 2011;106:2070–7.

IBD: Glucocorticoids

- Indication
 - **Active IBD:**
 - short term use (**weeks**) (avoid ADR)
 - Systemically or deliver to the site of action
 - **Not effective in maintenance of remission**
- Once remission has been induced, should be gradually tapered (over many weeks)
- Patient can be divided by response to steroid
 - Steroid-responsive: 40%
 - Steroid-dependent: 30-40%
 - Steroid-unresponsive: 15-20%

- Oral prednisolone
 - In active case, not responding to 5-ASA therapy
 - 20-60 mg/day
- Parenteral (IV) drugs:
 - In more severe disease
 - Hydrocortisone: 300 mg/day
 - Methylprednisolone: 40-60 mg/day

- **Adverse effects**
 - Weight gain
 - Fat redistribution
 - Hyperglycemia
 - Cataract
 - Emotional changes
 - Osteoporosis

Glucocorticoids: UC mild to moderate

Prednisolone is superior to 5-ASA for induction of remission in UC, but has significant side effects and should be reserved for patients with failure of response or who are intolerant to oral and/or rectal 5-ASA

- Budesonide MMX: 9 mg daily (1-3)
 - Pool analysis of CORE I and CORE II:
 - Compared oral budesonide MMX 9 mg/day and 6 mg/day with placebo in patients with mild to moderate left-sided and extensive UC.
 - week 8 combined clinical and endoscopic remission rates were 17.7% for budesonide MMX 9 mg versus 6.2% for placebo (p=0.0002) for left side UC.
 - 6 mg not better than placebo.
 - Ileal-release budesonide does not induce remission.
 - Inferior to placebo and 5-ASA
- ADR:
 - has a lower rate of systemic adverse effects than conventional corticosteroids (33% vs 55%),
 - not associated with adrenal suppression or a significant reduction in bone mineral density.

- Beclomethasone dipropionate:
 - 5 mg daily for 4 weeks, then alternate weekly for a further 4 weeks was shown to be non-inferior to prednisolone in efficacy.
 - Similar ADR
 - randomized controlled trial in 282 patients (4)
- In patients with active left-sided or extensive UC,
 - oral beclomethasone dipropionate 5 mg/day has been demonstrated as equivalent to 2.4 g 5-ASA,
 - more effective when added to 5-ASA compared with 5-ASA alone. (5-7)

1. Sandborn WJ, Travis S, Moro L, et al. Once-daily budesonide MMXR extended-release tablets induce remission in patients with mild to moderate ulcerative colitis: results from the CORE I study. *Gastroenterology* 2012;143:1218–26.
2. Travis SP, Danese S, Kupcinskas L, et al. Once-daily budesonide MMX in active, mild-to-moderate ulcerative colitis: results from the randomised CORE II study. *Gut* 2014;63:433–41.
3. Sherlock ME, MacDonald JK, Griffiths AM, et al. Oral budesonide for induction of remission in ulcerative colitis. *Cochrane Database Syst Rev* 2015;CD007698.

5. Van Assche G, Manguso F, Zibellini M, et al. Oral prolonged release beclomethasone dipropionate and prednisone in the treatment of active ulcerative colitis: results from a double-blind, randomized, parallel group study. *Am J Gastroenterol* 2015;110:708–15.
6. Campieri M, Adamo S, Valpiani D, et al. Oral beclomethasone dipropionate in the treatment of extensive and left-sided active ulcerative colitis: a multicentre randomised study. *Aliment Pharmacol Ther* 2003;17:1471–80.
7. Rizzello F, Gionchetti P, D'Arienzo A, et al. Oral beclomethasone dipropionate in the treatment of active ulcerative colitis: a double-blind placebo-controlled study. *Aliment Pharmacol Ther* 2002;16:1109–16.

Glucocorticoids: UC moderate to severe

- Oral corticosteroids are effective for the induction of remission (1-2)
 - in patients with moderate to severe UC refractory to 5-ASA,
 - in patient have responded to initial treatment with IV corticosteroids for acute severe disease.
- **Prednisolone: 40 mg daily (Baron's study(3))**
 - Single daily dosing is as effective as split-dosing and causes less adrenal suppression.
 - patients not responding after 2 weeks
 - considered for treatment escalation to biologics
 - admission to hospital
 - The dose should be tapered over 6–8 weeks.
- ADR:
 - Higher than 40-60mg/day not show benefit but increase ADR (4)
 - 50% of patients experience short-term corticosteroid-related adverse event
 - acne, edema, sleep and mood disturbance, glucose intolerance and dyspepsia

1. Lennard-Jones JE, Longmore AJ, Newell AC, et al. An assessment of prednisone, salazopyrin, and topical hydrocortisone hemisuccinate used as out-patient treatment for ulcerative colitis. Gut 1960;1:217–22.
2. Truelove SC, Watkinson G, Draper G. Comparison of corticosteroid and sulphasalazine therapy in ulcerative colitis. Br Med J 1962;2:1708–11.
3. Baron JH, Connell AM, Kanaghiniis TG, et al. Out-patient treatment of ulcerative colitis. Comparison between three doses of oral prednisone. Br Med J 1962;2:441–3.
4. Ford AC, Bernstein CN, Khan KJ, et al. Glucocorticosteroid therapy in inflammatory bowel disease: systematic review and meta-analysis. Am J Gastroenterol 2011;106:590–9.

Glucocorticoids: ileocecal CD

- **ileocecal CD-mild to moderate**
 - Ileal-release budesonide 9mg once daily for 8 weeks
- After remission
 - Budesonide should be tapered over 1-2 weeks
- ADR:
 - **Significant fewer of systemic adverse effects** than conventional corticosteroids
 - not associated with adrenal suppression or a significant reduction in bone mineral density.

- **Budesonide MMX: 9 mg daily (1-3)**
 - Meta analysis (1-3)
 - **budesonide 9 mg/day over placebo** in inducing remission
 - A once-daily 9 mg dose is as effective as 3 mg three times daily
 - In a randomised double-blind trial; mild-to-moderate ileocecal Crohn's disease (4)
 - Induction remission; ileal-release budesonide 9 mg once daily for 8 weeks was **as effective as prednisolone** (CDAI score <150) (40 mg daily tapering to 5 mg at 8 weeks)
 - In **more severe disease** (CDAI > 300)
 - budesonide is inferior to prednisolone (RR 0.52, 95% CI 0.28 to 0.95) (2)

CDAI: The Crohn's Disease Activity Index

1. Moja L, Danese S, Fiorino G, et al. Systematic review with network meta-analysis: comparative efficacy and safety of budesonide and mesalazine (mesalamine) for Crohn's disease. Aliment Pharmacol Ther 2015;41:1055-65.
2. Rezaie A, Kuenzig ME, Benchimol EL, et al. Budesonide for induction of remission in Crohn's disease. Cochrane Database Syst Rev 2015(6):CD000296.

3. Coward S, Kuenzig ME, Hazlewood G, et al. Comparative effectiveness of mesalamine, sulfasalazine, corticosteroids, and budesonide for the induction of remission in Crohn's disease: a Bayesian network meta-analysis. Inflamm Bowel Dis 2017;23:461-72.
4. Bar-Meir S, Chowers Y, Lavy A, et al. Budesonide versus prednisone in the treatment of active Crohn's disease. The Israeli Budesonide Study Group. Gastroenterology 1998;115:835-40.

Glucocorticoids: colonic CD

- Colonic Crohn's disease (mild, moderate, severe) + severe ileocolonic CD
 - Systemic corticosteroids such as prednisolone are effective in inducing remission (1-3).
 - Recommend treated with an 8-week course of systemic corticosteroid
 - A starting dose of 40 mg tapering by 5 mg weekly.
 - Ileal-release budesonide
 - benefit in CD affecting the proximal colon (4)
 - no evidence of benefit in CD at more distal colonic inflammation
- There are no trials of colonic-release budesonide-MMX in Crohn's disease

Treatment option in IBD

Biologic, immunosuppressive, immunomodulator therapy

- Thiopurine
- Infliximab
- Adalimumab
- Golimumab
- Vedolizumab
- Tofacitinib
- Ustekinumab
- Methotrexate

Factors to consider in choosing a biologic, immunosuppressive or immunomodulator therapy

- ▶▶ Route of administration (oral, subcutaneous, intramuscular, intravenous)
- ▶▶ Speed of response to induction therapy (consider need for bridging therapy)
- ▶▶ Potential immunogenicity and need for combination therapy
- ▶▶ Side effects including cancer risk
- ▶▶ Persistence (continuing drug without loss of response after initial improvement)
- ▶▶ Availability of infusion facilities and therapeutic drug monitoring
- ▶▶ Overall cost (including drug delivery and monitoring)

Thiopurine derivatives

- **Thiopurine derivative**
 - **Mercaptopurine (6-MP)**
 - **Azathioprine**

- **Purine analog:**
 - both are prodrugs: Azathioprine, 6-mercaptopurine
- **Mechanisms of action**
 - Impair purine biosynthesis; reduce IBD-associated GI inflammation
 - Inhibit cell proliferation; immune cell
- **Use**
 - Maintenance of remission in UC
 - Maintenance of remission in CD

- Meta-analysis of 3-RCT (**thiopurine** VS placebo) (1-2)
 - RR 0.6, 95% CI 0.37 to 0.95
 - OR was 2.59 (95% CI 1.26 to 5.3)
 - absolute risk reduction was 23%
 - NNT to prevent one recurrence was 5
 - **Reduce the long-term use of corticosteroid**
 - Slow onset of action (3-12 months); limited use in active IBD
- Cochrane review from four maintenance (3)
 - **azathioprine** versus placebo
 - benefit of azathioprine over placebo failure (44% vs 65%)
 - RR 0.68, 95% CI 0.54 to 0.86

thiopurine derivative: ADR

- Drug interaction
- Inhibition of xanthine oxidase by allopurinol divert metabolism toward production of 6-thioguanine analogs
- Augments the risk of immunosuppression
- Patient on mercaptopurine should be warned about serious interactions with medications used to treat gout and hyperuricemia
- If the patient is already taking allopurinol
 - Dose decreased to 25% of standard dose

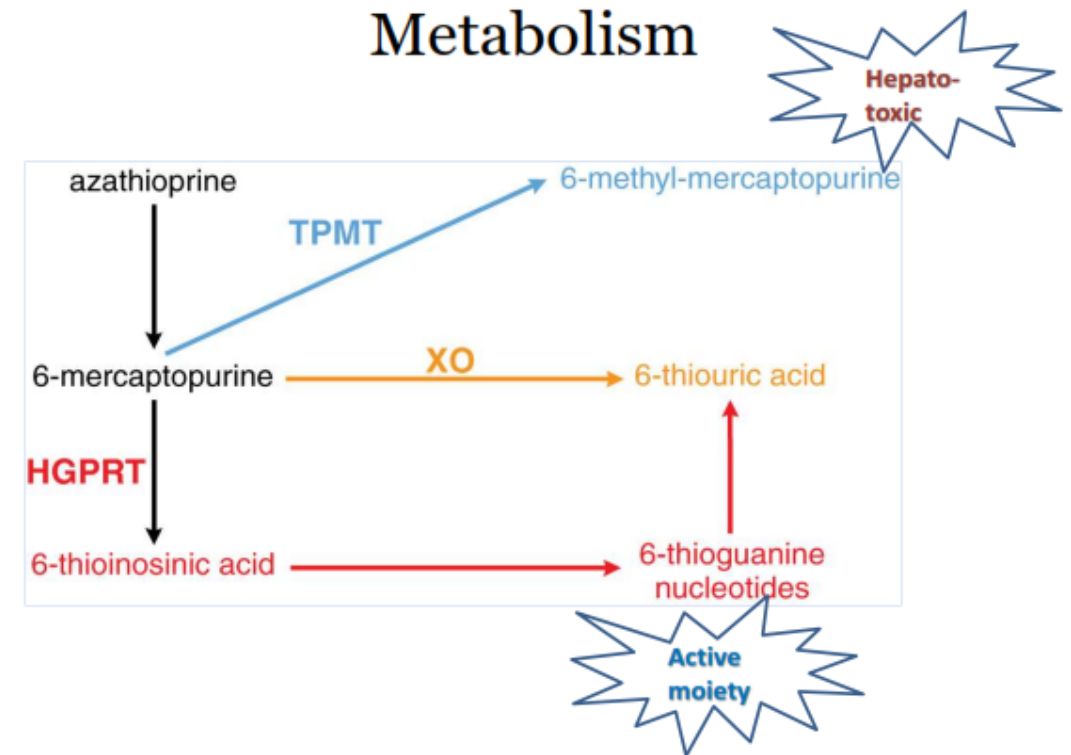
Thiopurine methyl transferase (TPMT)

Patient with **minimal TPMT activity**

- Metabolism shifted toward 6-thioguanine nucleotides
 - Bone marrow suppression
 - Weekly monitoring CBC

Patient with **rapid metabolizer**

- Metabolism shifted toward 6-MMP
 - Hepatotoxicity
 - Reduced therapeutic response
 - LFT monitoring done



Infliximab: anti TNF- α in UC

- Chimeric IgG1 antibody (25% mouse, 75% human)
- Mechanism of action
 - Bind to free-from and membrane-bound TNF-alpha
- Use
 - **Moderate to sever UC**
 - 5mg/kg i.v. infusion

- ACT1 and ACT2 clinical trials,
 - **Active UC** (corticosteroids and/or thiopurines (and/or 5-ASA (ACT2)) (1)
 - Plus 5 or 10 mg/kg infliximab at week 0,2,6, placebo
 - **Week8, similar clinical response in both dose**
 - Clinical response 5 mg/kg better than placebo (67% VS 33% response)
 - 5mg/kg
 - **Clinical remission rate at Week30** were 30% for 5 mg/kg (13% for placebo), sustained to week54.
 - **Corticosteroid-free remission rate at Week30**, were 22% for 5mg/kg, sustain to week 54
- Cohort studies (2-3)
 - Infliximab for patients **with corticosteroid-dependent UC**
 - steroid-free remission rates of 47% at 1 year
 - 77% colectomy-free at median 41.5 months
 - better outcomes for those on combination therapy with thiopurines or thiopurine-naïve

Infliximab: anti TNF- α in CD

- Chimeric IgG1 antibody (25% mouse, 75% human)
- Mechanism of action
 - Bind to free-from and membrane-bound TNF-alpha
- Use
 - **Moderate to sever CD**
 - 5mg/kg i.v. infusion
 - **Recommended combination therapy with thiopurine for induction and maintenance**

- ACCENT 1 clinical trials,
 - Active luminal CD (573 patients) (1)
 - 5 mg/kg infliximab at week 0,2,6, or placebo
 - **Week2**; 58% response to initial infusion
 - Week30; 39% maintenance, 45% remission and sustain to week 54
- Cohort studies (614 patients) (2)
 - Infliximab for patients with CD
 - 89.1% clinical improvement at initial infusion
 - 63.4% sustained clinical benefit
- **Combination therapy with immunomodulator**
 - SONIC (3); infliximab combination with azathioprine was superior clinical outcomes
 - PANTS (4); immunomodulator decreased the risk of immunogenicity in infliximab use

1. Hanauer SB, Feagan BG, Lichtenstein GR, et al. Maintenance infliximab for Crohn's disease: the ACCENT I randomised trial. Lancet 2002;359:1541-9.
2. Schnitzler F, Fidder H, Ferrante M, et al. Long-term outcome of treatment with infliximab in 614 patients with Crohn's disease: results from a single-centre cohort. Gut 2009;58:492-500.
3. Colombel JF, Sandborn WJ, Reinisch W, et al. Infliximab, azathioprine, or combination therapy for Crohn's disease. N Engl J Med 2010;362:1383-95.
4. Kennedy NA, Heap GA, Green HD, et al. Predictors of anti-TNF treatment failure in anti-TNF-naive patients with active luminal Crohn's disease: a prospective, multicentre, cohort study. Lancet Gastroenterol Hepatol 2019;4:341-53.

Adalimumab: anti TNF- α in UC

- Recombinant human monoclonal IgG1 antibody
- Mechanisms of action
 - Binds TNF-alpha and neutralizes its function by blocking the interaction between TNF and its cell-surface receptor
- Use: **moderate to severe UC**
 - Injected subcutaneously
- **Less immunogenic than infliximab**

- ULTRA1 and ULTRA2 trials (moderate to severe UC) (1-2)
 - Adalimumab 160 mg at week 0, 80 mg at week 2, followed by 40 mg fortnightly
 - Week8; Remission 19-21% (placebo 9-11%)
 - Week52; ULTRA2 maintenance study
 - Maintain remission 22% (placebo12%)
 - Corticosteroid-free 14% (placebo6%)
- ULTRA3 (3)
 - 25% remained clinical remission at 4 years (fortnightly or weekly use)
- Real world data (4-5)
 - **Adalimumab comparable efficacy to infliximab** in moderate to severe UC patients naïve to anti-TNF therapy

1. Reinisch W, Sandborn WJ, Hommes DW, et al. Adalimumab for induction of clinical remission in moderately to severely active ulcerative colitis: results of a randomized controlled trial. Gut 2011;60:780-7.
2. Sandborn WJ, van Assche G, Reinisch W, et al. Adalimumab induces and maintains clinical remission in patients with moderate-to-severe ulcerative colitis. Gastroenterology 2012;142:257-65.
3. Colombel J-F, Sandborn WJ, Ghosh S, et al. Four-year maintenance treatment with adalimumab in patients with moderately to severely active ulcerative colitis: Data from ULTRA 1, 2, and 3. Am J Gastroenterol 2014;109:1771-80.
4. Sandborn WJ, Sakuraba A, Wang A, et al. Comparison of real-world outcomes of adalimumab and infliximab for patients with ulcerative colitis in the United States. Curr Med Res Opin 2016;32:1233-41.
5. Armuzzi A, Taxonera C, Panes J, et al. P365 Real-world effectiveness of adalimumab in patients with ulcerative colitis. J Crohns Colitis 2016;10(Suppl 1):S280-2.

Adalimumab: anti TNF- α in CD

- Recombinant human monoclonal IgG1 antibody
- Mechanisms of action
 - Binds TNF-alpha and neutralizes its function by blocking the interaction between TNF and its cell-surface receptor
- Use: **moderate to severe UC**
 - Injected subcutaneously
 - 160 mg then 80 mg at week2 for induction therapy
- **Less immunogenic than infliximab**

CLASSIC I; moderate to severe CD (1)

- 160 mg initial then 80 mg at week2
 - 36% achieve CDAI < 150 (12% placebo)

CHARM; maintenance therapy study (2)

- 80 mg initial then at week 2 to responder to induction therapy
 - Week 56; 36-41% response (placebo 12%)

GAIN; failed to infliximab (3)

- Adalimumab showed efficacy in these patients

Golimumab: anti TNF- α in UC

- Recombinant human monoclonal antibody
- Mechanisms of action
 - Binds TNF-alpha (both soluble and transmembrane bioactive form)
 - Prevent the interaction between TNF and its cell-surface receptor
- Use: moderate to severe UC
 - Injected subcutaneously

The PURSUIT SC trial (1)

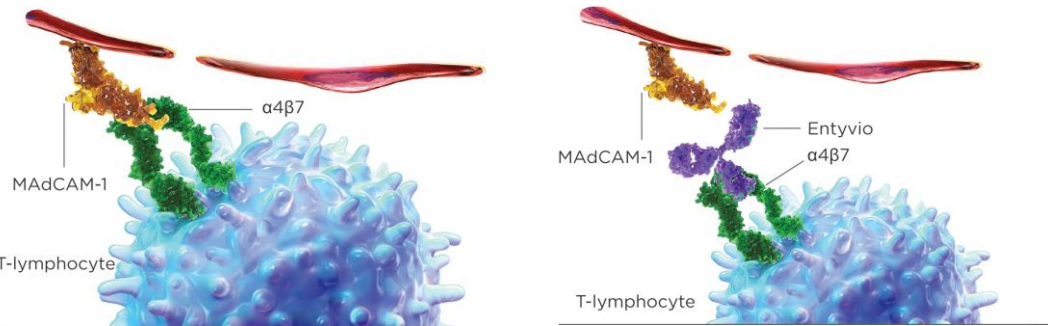
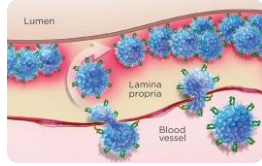
- a dose-finding phase
- 1064 patients with UC (anti-TNF naïve)
 - failed to respond to 5-ASA, oral corticosteroids, azathioprine or mercaptopurine.
 - steroid-dependent
- Week6: clinical response
 - 51% on 200mg/100mg
 - 54.9% on 400mg/200mg
 - 30.3% placebo
- Week6: clinical remission
 - 17.8% on 200mg/100mg
 - 17.9% on 400mg/200mg
 - 6.4% placebo

Network meta-analyses (2-5)

- conflicting data on the relative efficacy of infliximab, adalimumab and golimumab

Vedolizumab: $\alpha 4\beta 7$ integrin antagonist

- Recombinant monoclonal antibody
- Mechanisms of action
 - Binds $\alpha 4\beta 7$ integrin
 - blocking the entry of inflammation-stimulating lymphocytes
- Use: moderate to severe UC
 - Intravenous
- Recommend for anti-TNF failed (induction, maintenance)



<https://www.entyviohcp.com/mechanism-of-action>

GEMINI I clinical trial; active UC (1-2)

- IV 300 mg at day1 and 15
 - Week6; clinical response 53.1% vs 26.3% placebo
 - Week6; clinical remission 23.1% vs 6.6% placebo
 - week52; maintenance 60.7% vs. 26.6% placebo
 - no clinically important differences in safety vs placebo

VICTORY consortium (3)

- experienced with anti-TNF therapy reduced VD efficacy

European retrospective study; anti-TNF naïve patients (4)

- 50 Crohn's disease and 134 UC
 - Week14
 - Response 79.1%
 - Clinical remission 39.5%
 - Steroid –free remission 36.6%

VARSIITY; head-to-head in IBD (vedolizumab vs. adalimumab) (5)

- 769 moderate to severe active IBD, double blind, double dummy
 - Week 52; clinical remission 31.3% VD, 22.5% AD (p=0.006)
 - Week 52; mucosal healing 39.7% VD, 27.7% AD (p=0.0005)
 - Week52; steroid-free remission not different

These data support for vedolizumab as a first-line biologic option for UC failing conventional therapy.

1. Feagan BG, Rutgeerts P, Sands BE, et al. Vedolizumab as induction and maintenance therapy for ulcerative colitis. N Engl J Med Overseas Ed 2013;369:699–710.
2. Feagan BG, Rubin DT, Danese S, et al. Efficacy of vedolizumab induction and maintenance therapy in patients with ulcerative colitis, regardless of prior exposure to tumor necrosis factor antagonists. Clin Gastroenterol Hepatol 2017;15:229–39.
3. Narula N, Peerani F, Meserve J, et al. Open: Vedolizumab for ulcerative colitis: treatment outcomes from the VICTORY consortium. Am J Gastroenterol 2018;113:1345–54.
4. Kopylov U, Verstockt B, Biedermann L, et al. Effectiveness and safety of vedolizumab in anti-TNF-naïve patients with inflammatory bowel disease—a multicenter retrospective European study. Inflamm Bowel Dis 2018;24:2442–51.
5. Sands BE, Peyrin-Biroulet L, Loftus EV, et al. 416a – Vedolizumab shows superior efficacy versus adalimumab: results of Varsity—the first head-to-head study of biologic therapy for moderate-to-severe ulcerative colitis. Gastroenterology 2019;156:S-81.

Tofacitinib (JAK inhibitor)

- **Small molecule inhibitor**
- Mechanisms of action
 - Prevent phosphorylation and activation of STAT
 - Inhibit immune cell function
- Use: moderate to severe UC (oral)
- **No immunogenicity**
- Approved for the treatment (induction and maintenance) of moderate to severely active UC where conventional treatments or anti-TNF have not been tolerated, or have failed
 - **High dose should not be used in high risk of pulmonary embolus**

Phase2 dose ranging induction study: (moderate to severe UC) (1)

- Significance remission after 8 weeks in high dose (10, 15 mg BID)

The OCTAVE-1 and OCTAVE-2: (2)

- Confirm efficacy of 10mg twice daily in active UC
- Week8: remission; 18.5% (OCTAVE-1), 16.6% (OCTAVE-2)
- Patient with anti-TNF naïve
 - Week8: remission; 23.7%

OCTIVE-SUSTAIN (pooled patients from both trials to RCT) (3)

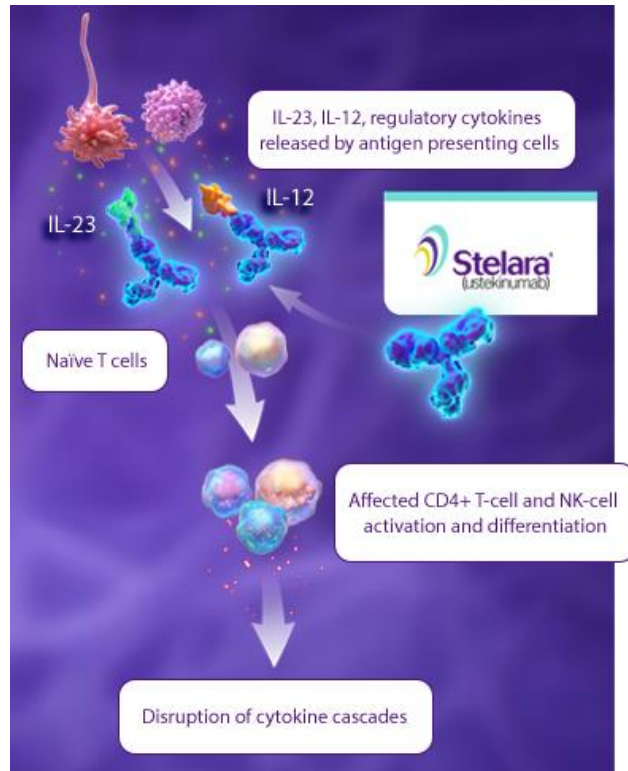
- After 1 years; remission achieved
 - 34.4% in 5 mg, 40.6% in 10 mg, 11.1% in placebo
- Retreatment is effective after a break in therapy (4)

ADR: did not differ from placebo group (required further data)

- Increase risk of infection
 - **Zoster vaccination** (prevent herpes zoster) before 4 weeks
 - Before starting therapy ; aged >70 years and >50 years at high risk

Ustekinumab: Interleukin antagonist in UC

- Monoclonal antibody-IL12/23p40
- Mechanisms of action
 - Inhibit IL-12, IL-23
 - Inhibit the activation of T-cell
- **Required more study in UC**



The UNIFI trial, (moderate to severe UC)

- Adults **failed to** corticosteroids, immunomodulators, anti-TNF therapy, vedolizumab (1)
- 130mg or 6mg/kg) ustekinumab, IV
 - Week8; clinical remission
 - 15.6% on 130 mg, 15.5% on 6 mg/kg
 - Week8; Endoscopic healing
 - 26.3% on 130mg, 27% on 6 mg/kg and
 - Week8; clinical response
 - 51.3% on 130 mg, 61.8% on 6 mg/kg

There was no difference in adverse events compared with placebo, with no malignancies, opportunistic infections or TB reported.

In the maintenance continuation study, 523 patients with clinical response at week 8 were re-randomised (2)

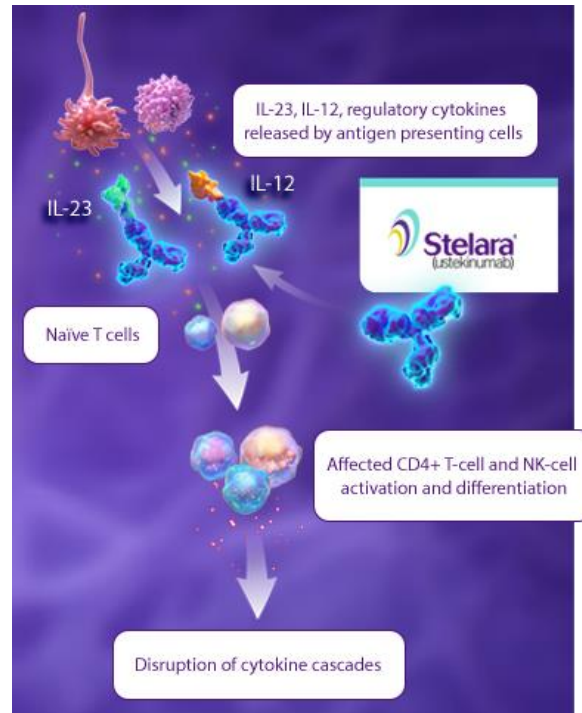
- week 44 remission rates of 24% (placebo), 38.4% (8-weekly dosing) and 43.8% (12-weekly dosing)
- Effects were seen in both anti-TNF naïve and anti-TNF refractory patients (**favor**)

does not have licensing or approval for its use in UC

1. Sands BE, Sandborn WJ, Panaccione R, et al. LB01 Safety and efficacy of ustekinumab induction therapy in patients with moderate to severe ulcerative colitis: results from the Phase 3 UNIFI study. United Eur Gastroenterol J 2018;6:1586–97.
2. 154 Sandborn WJ, Sands BE, Panaccione R, et al. OP37 Efficacy and safety of ustekinumab as maintenance therapy in ulcerative colitis: Week 44 results from UNIFI. J Crohns Colitis 2019;13(Suppl 1):S025–6.

Ustekinumab: Interleukin antagonist in CD

- Monoclonal antibody-IL12/23p40
- Mechanisms of action
 - Inhibit IL-12, IL-23
 - Inhibit the activation of T-cell
- **Recommend; ustekinumab can be used in induction and maintenance of CD; both anti-TNF naïve and anti-TNF failed**



UNITI-1 trial, (moderate to severe CD)

- Adults **failed to** anti-TNF therapy (1)
- 130 mg or 6mg/kg ustekinumab, IV
 - Week8; clinical response
 - 33.5% on 130 mg, 37.8 % on 6 mg/kg

UNITI-2 trial, (moderate to severe CD) (1)

- Adult; no previous failed to anti-TNF therapy
- 130 mg or 6mg/kg ustekinumab, IV
 - Week8; clinical response
 - 47.4 % on 130mg, 57.9% on 6 mg/kg

IM-UNITI, (maintenance)(90 mg subcutaneous) (1)

- Week 44; remission
 - 48.8-53.1% of ustekinumab
 - 41.1% of anti-TNF refractory patient

Immunosuppressants: Methotrexate in UC

- Mechanism of action
 - Impaired DNA synthesis due to Dihydrofolate reductase inhibition
 - Decreased IL-1 production
- Use
 - **No role in induction or maintenance in UC**

Cochrane review 2015: methotrexate compare with 5-ASA, sulfasalazine, mercaptopurine, placebo (1)

- Result not support to use in maintenance UC

European double-blind RT: methotrexate VS placebo (2)

- Methotrexate was not superior to placebo

MERIT: active UC then maintenance (3)

Methotrexate + prednisolone tapering 12 weeks then 84 patients were continued methotrexate VS placebo

- Week 48: no different outcome between Methotrexate Vs placebo

Immunosuppressants: Methotrexate in CD

- Mechanism of action

- Impaired DNA synthesis due to Dihydrofolate reductase inhibition
- Decreased IL-1 production

- Use

- Recommend; methotrexate may be used for the maintenance of remission of CD, dose should be at least 15 mg weekly
- Subcutaneous administration has better bioavailability than oral in high dose

A meta-analysis and systematic review (1)

- Benefit of methotrexate in maintenance therapy
 - OR 2.4 (95% CI 1.1 to 4.8)

Route of administration

- Oral methotrexate has reduced and extremely variable bioavailability than parental administration (2-3)
- Subcutaneous and intramuscular methotrexate have similar bioavailability to parental administration (4)

1. Hazlewood GS, Rezaie A, Borman M, et al. Comparative effectiveness of immunosuppressants and biologics for inducing and maintaining remission in Crohn's disease: a network meta-analysis. *Gastroenterology* 2015;148:344–54.
2. Kurnik D, Loebstein R, Fishbein E, et al. Bioavailability of oral vs. subcutaneous low-dose methotrexate in patients with Crohn's disease. *Aliment Pharmacol Ther* 2003;18:57–63.
3. Wilson A, Patel V, Chande N, et al. Pharmacokinetic profiles for oral and subcutaneous methotrexate in patients with Crohn's disease. *Aliment Pharmacol Ther* 2013;37:340–5.
4. Brooks PJ, Spruill WJ, Parish RC, et al. Pharmacokinetics of methotrexate administered by intramuscular and subcutaneous injections in patients with rheumatoid arthritis. *Arthritis Rheum* 1990;33:91–4.

Nutritional therapy: Crohn's disease

- **Exclusive Enteral Nutrition (EEN)**

- Suggest to use for inducing remission in mild to moderate CD
 - Primary treatment option in pediatric
 - Avoidance of corticosteroid (pediatric)
 - Strictly adherence
 - Symptoms relief within 10 days
 - Continue up to 8 weeks for mucosal healing

- Dietary intervention has no role in UC

Exclusive Enteral Nutrition (EEN)

- Meta-analysis (1-2)
 - Induce remission as effective as corticosteroid (mild to moderate) in pediatric patients (73% intention to treat), not in adult

Clinical features: extraintestinal manifestation

- **Joint**
 - Sacroiliitis
 - Ankylosing spondylitis
- **Ocular**
 - Episcleritis
 - Uveitis
 - Iritis (blurred vision)
 - Eye pain
 - photophobia
- **Skin**
 - Pyoderma gangrenosum
 - Erythema nodosum
- **Kidney**
 - Nephrolithiasis
 - Oxalate stone (more common in CD)
 - Uric acid-containing stone (more common in UC)
- **Liver**
 - CD: increased incidence of gallstone formation
 - UC: sclerosing cholangitis, cholangiocarcinoma

